

Module-4 (Assignment Day 2)

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).



5 Common Desktop PC Hardware Issues

1. **No Display on Monitor** -Symptom: Blank screen, no display, or no POST beep.
2. **RAM Failure** - Symptom: PC does not boot, repeated beeps, or frequent crashes.
3. **Hard Disk/SSD Failure** -Symptom: PC becomes very slow, operating system does not load, or storage is not detected.
4. **Overheating** - Symptom: PC shuts down suddenly, becomes very hot, or runs slowly.
5. **Power Supply (SMPS) Failure** -Symptom: PC does not turn on, no fans or lights, or system suddenly switches off.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.



Laptop Not Charging – Step-by-Step Troubleshooting

1. **Check the Power Socket** - Plug another device into the socket to confirm that the socket is working properly.
2. **Inspect the Charger** - Check the charger, cable, and plug for cuts, burns, loose wires, or other physical damage.
3. **Check the Charging Connector** - Make sure the charger pin is firmly connected to the laptop and the port is not loose, bent, or damaged.
4. **Check the Charger Output** - If a basic multimeter is available, carefully check the charger's DC output voltage and compare it with the voltage printed on the charger label.
5. **Try Another Compatible Charger** - Use a known-working charger with the correct voltage, connector, and suitable current rating to determine whether the original charger is faulty.
6. **Perform a Power Reset** - Disconnect the charger, switch off the laptop, remove the battery if it is removable, hold the power button for about 15–20 seconds, then reconnect the charger and try again.
7. **Check the Battery** - Look for swelling, leakage, or physical damage. Do not use a swollen or damaged battery.
8. **Check the Result** - If the laptop works with the charger but does not charge the battery, the battery or charging circuit may be faulty. If it does not power on even with a known-good charger, professional hardware repair may be required.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).



Multimeter Test of a USB Phone Charger

Tools used: Digital multimeter and USB phone charger.

1. Set the multimeter to DC Voltage (V_{DC}).
2. Plug the charger into a working wall socket, without connecting it to the phone.
3. Connect the multimeter red probe to the USB +5V/VBUS contact and the black probe to GND.
4. The expected USB voltage is approximately 5V DC. For the common USB 5V range used in this test, 4.75V–5.25V is considered acceptable
5. Sample recorded reading: 5.08V DC.
6. Result: 5.08V is between 4.75V and 5.25V, so the charger passes this basic voltage check.

Record for assignment:

Measured voltage = 5.08V DC

Expected range = 4.75V–5.25V

Conclusion = Voltage is within the expected range.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.



Testing PSU 12V Rail with a Multimeter

1. Switch off and unplug the PC from the wall.
2. Disconnect the ATX power connector from the motherboard.
3. Set the multimeter to DC Voltage (V =).
4. Put the black probe on a black (GND) wire and the red probe on the yellow (+12V) wire.
5. Record the reading.

Measured voltage: 12.1V

Result: 12.1V is within the normal range.

If the voltage is below 11.5V: It could indicate a faulty or weak power supply (SMPS/PSU), which may cause random shutdowns.